Band-1 Hadamard Order 5716 — Miyamoto (Paley-I H_1428 + C_1430 over GF(1429)) + Paley-II SPK; IN SPK via Paley-II; Gap 2052 Still OPEN

H_1428/C_1430/H_5716 Miyamoto PASS; H_5716 Paley-II PASS (q=2857); 5716 IN SPK via Paley-II; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_5476_Miyamoto`; `2026-09-15_Band1_Hadamard_5188_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_5716_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1428 and 5716 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with preferred Paley-I seed H_{1428} = Paley-I($q = 1427$) and symmetric Paley conference C_{1430} over **GF(**1429**)** (prime field, $q = 1429 \equiv 1 \pmod 4$). Double path for H_{1428} is **SKIP** ($q = 713 = 23 \cdot 31$ not prime). Classical SPK path **Paley-II** with $q = 2857$ (prime $\equiv 1 \pmod 4$) is **constructed and verified** → order **5716 IN SPK via Paley-II** (same honesty pattern as 5188). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Paley-I for 5716 FAIL ($q = 5715$ not prime); Kronecker FAIL (4×1429). `clay_closed = false`.

Abstract

Finite constructions of orders **1428** and **5716** via Miyamoto + Paley-II.

- H_{1428} = Paley-I ($q = 1427$) — preferred; Double SKIP ($q = 713$ not prime)
- C_{1430} = Paley conference over GF(1429) prime field
- H_{5716} = Miyamoto(H_{1428}, C_{1430})
- H_{5716} also = Paley-II ($q = 2857$) — classical SPK path

SPK: 5716 IN SPK via Paley-II (q=2857). Seed 1428 enters SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_5716.py`

Results. overall_pass=true; units 64/64; wall 304.47s; all max|HH −nI|=0.

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Arithmetic

- Target $n = 5716 = 4 \cdot 1429$; 1429 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1428} and C_{1430} (Paley conference over prime field 1429).
- Seed preferred: Paley-I $q = 1427$ (prime $\equiv 3 \pmod 4$).
- Double SKIP: half=714, $q = 713$ not prime.
- SPK: Paley-I $q = 5715$ **FAIL**; Paley-II $q = 2857$ **PASS** → **IN SPK**; Kronecker **FAIL**.

Constructions (verified)

Object	Method	max|HH −nI|	Wall (s)
H_1428 Paley-I	Paley-I q=1427 (preferred)	0	2.4082
H_1428 Double	SKIP q=713 not prime	—	—
C_1430	Paley conf GF(1429) q=1429	0	2.2834
H_5716 Miyamoto	Miyamoto(H_1428, C_1430)	0	146.2347
H_5716 Paley-I	SKIP q=5715 not prime	—	—
H_5716 Paley-II	Paley-II q=2857 (SPK path)	0	148.5807

SPK closure note

Seeds include prior SPK ships through **5444** + **684/1368** (from 5476 run) + verified **1428**; **5716** folded via Paley-II; **exclude** Miyamoto-only bonuses including **5476/5284**. Known count 1201. Next constructive gap **2052** (OPEN).

Quantity	Value
5716 in SPK	**True** (via Paley-II q=2857)
1428 in SPK	**True** (Paley-I)
5476 in SPK	**False** (prior Miyamoto-only)
5444 in SPK	**True** (prior Paley-I)
Next constructive gap	**2052** (OPEN)

Honesty

- clay_closed = false
- GATE-04 = LOCKED
- Hadamard conjecture = OPEN
- Gap 2052 = OPEN (method triage / non-finding ONLY)
- 5716_in_spk = true (via Paley-II)
- 5716_miyamoto_only = false

References (classical; no fake DOIs)

- Paley type I/II constructions (prime / prime-power fields).
- Symmetric Paley conference matrices for $q \equiv 1 \pmod 4$.
- Miyamoto construction of Hadamard matrices of order $4q$ from H_{q-1} + conference of order $q+1$.
- SPK = Sylvester Paley I/II Kronecker; Miyamoto not folded into SPK.